



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE
NAME

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CENTRE
NUMBER

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CLASS

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INDEX
NUMBER

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H2 BIOLOGY

9744/04

Paper 4 Practical

28 Aug 2017

Candidates answer on the Question Paper.

2 hours 30 minutes

Additional Materials: As listed in the Apparatus and Materials List.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your class, index number and name on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You will lose marks if you do not show your working, or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift	
Laboratory	
For Examiner's Use	
1	/ 22
2	/ 19
3	/ 14
Total	/ 55

Answer **all** questions.

Question 1

In this question, you will investigate the effect of glucose and sucrose on the rate of respiration in yeast cells.

You are provided with:

- 40 cm³ yeast cell suspension, labelled **Y**
- 25 cm³ blue alkaline solution, labelled **C**
- 25 cm³ hydrochloric acid, labelled **H**
- 20 cm³ of 0.2 mol dm⁻³ glucose solution, labelled **G**
- 20 cm³ of 0.2 mol dm⁻³ sucrose solution, labelled **S**
- 20 cm³ of distilled water, labelled **W**

*Read through steps **1** to **12** and prepare a table to record your results in step **(c)**, before starting the investigation.*

Stage 1

Perform the following steps to activate the yeast cells in **Y** to prepare for **stage 3**

- 1** Label 3 boiling tubes **B1**, **B2**, and **B3**. Add 10 cm³ of **Y** into each of the boiling tubes.
- 2** Add 10 cm³ of **G**, **S**, **W** into boiling tubes **B1**, **B2**, and **B3**, respectively.
- 3** Start the stopwatch. Leave the tubes aside for at least 15 minutes, before beginning on **stage 3**. During this time, you should attempt the rest of the question.

Stage 2

C, an alkaline solution, is blue because it contains an indicator.

Carbon dioxide reacts with **C** when bubbled into it. When enough carbon dioxide reacts with **C**, the indicator will change from blue to yellow (even if the mixture is cloudy). This is the end-point.

If only a small volume of carbon dioxide is bubbled into **C** then the indicator will remain blue. The end-point can then be reached by adding hydrochloric acid, **H**, slowly until the indicator turns from blue to yellow. The volume of **H** is then recorded.

The volume of **H** added to reach the end-point indicates how much carbon dioxide had reacted with **C**. The lesser the amount of carbon dioxide is bubbled into **C**, the more volume of **H** will need to be added to **C** to get the end-point.

You are required to find the volume of **H** needed to reach the end-point when **no** carbon dioxide has been bubbled into **C**.

- 4 Put 5 cm³ of **C** into a test-tube.
- 5 Use a 6 cm³ syringe, containing 3 cm³ of **H**, to put drops of **H** into **C** as shown in Fig. 1.1. Mix well as you add **H**, until the end-point is reached. You may need to fill the syringe again.

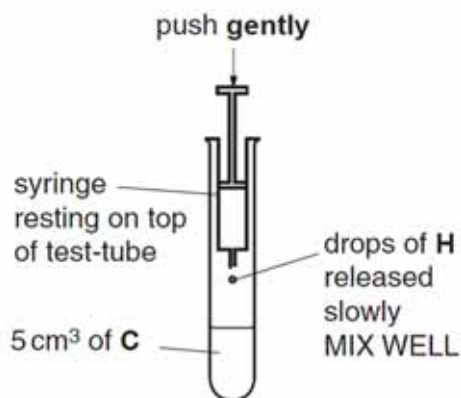


Fig. 1.1

- (a) Record the volume of **H** needed to reach the end-point. [1]

Volume of **H** needed = _____ cm³

Stage 3

You are required to investigate the effect of glucose and sucrose on the release of carbon dioxide from a yeast cell suspension using apparatus set up as in Fig.1.2, and the tubes prepared in **stage 1**.

Yeast cells release carbon dioxide from some of their metabolic reactions when provided with respiratory substrates. Tube **B1** contains glucose as the respiratory substrate, whilst tube **B2** contains sucrose.

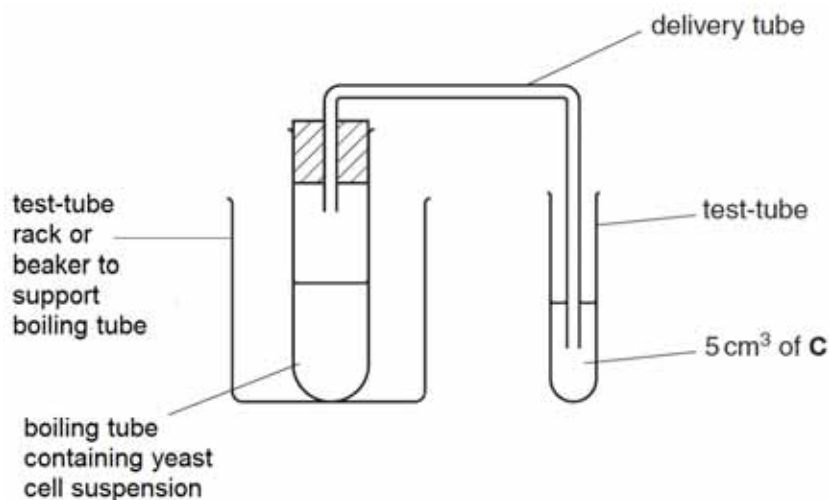


Fig.1.2

- (b) State which sugar is expected to lead to a higher rate of carbon dioxide release by yeast cells. Explain your answer. [2]

Before proceeding with steps 6-12, ensure that 15 minutes have elapsed since stage 1.

- 6** Prepare and maintain a water bath at 40°C.
- 7** Label 3 test-tubes **T1**, **T2** and **T3**. Add 5 cm³ of **C** into each of the three test-tubes.
- 8** Place tube **B1** in the water bath and allow it to equilibrate for 2 minutes.
- 9** Put the bung containing the delivery tube into tube **B1**, as shown in Fig. 1.2. Ensure that the seal is airtight.
- 10** Put the end of the delivery tube into solution **C** in **T1**. Using a stopwatch, allow the carbon dioxide produced by the yeast to bubble into **C** for 2 minutes.
- 11** After removing the delivery tube, repeat **step 5** to determine the volume of **H** needed to reach end-point and record your result in (c). If **C** has already reached end-point, record '0'.
- 12** Repeat steps **8** to **11** for the tubes **B2** and **B3**.
- (c) Record your results for each tube in a suitable form in the space below. [4]

- (d) Discuss what these results suggest about the relationship predicted in part (b). [2]

The addition of 1 cm³ of **H** into **C** is equivalent to 2.2 mg of carbon dioxide.

- (e) Calculate the exact amount of carbon dioxide produced by the yeast suspension from boiling tubes **B1** and **B2** during the experiment. Show your working clearly. [2]

B1 _____ mg

B2 _____ mg

- (f) State the purpose of including boiling tube **B3** in the investigation. [1]

- (g) Identify **one** significant source of error and suggest a modification to the method that would overcome this. [2]

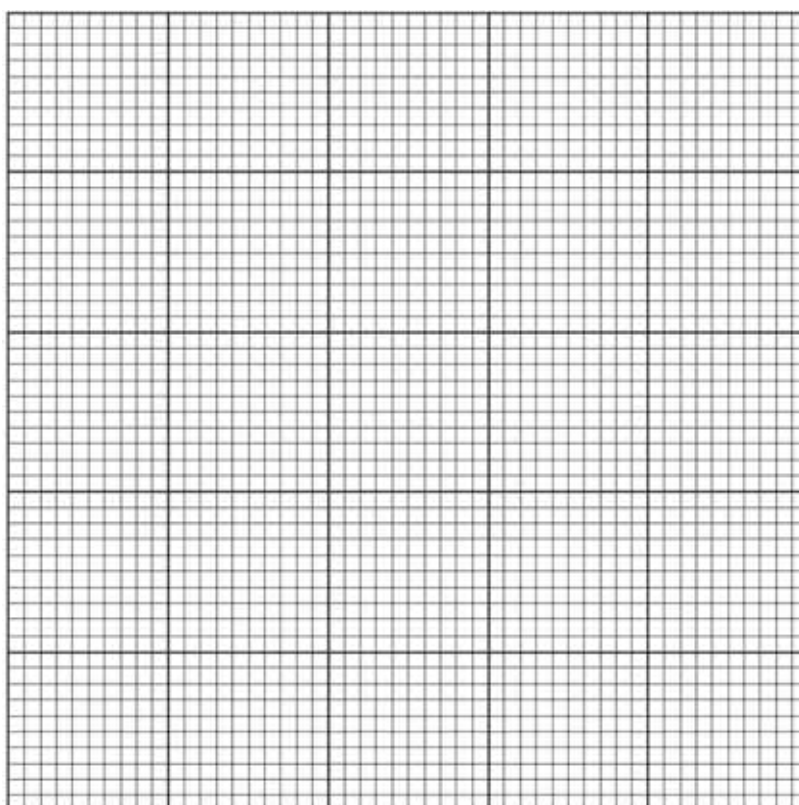
In a separate study, a student used a modified version of the apparatus shown in Fig.1.2 to investigate the effect of temperature on the release of carbon dioxide from a yeast cell suspension. The results of the study are presented in Table 1.1.

Table 1.1

Temperature / °C	Volume of carbon dioxide evolved / cm ³
25.0	93
30.0	126
35.0	160
40.0	180
45.0	156

(h) Use the grid provided to plot the data given in Table 1.1.

[4]



(i) Explain the relationship between the variables in yeast respiration.

[4]

[Total: 22]

Question 2

During this question you will require access to a microscope.

You are required to investigate the effects of potassium nitrate and lead nitrate solutions on cells of the plant material with which you have been supplied. Peel off one or two strips of epidermis from the most deeply pigmented areas of the plant tissue. Remove as little of the underlying tissue as possible. Cut the epidermis so that you have two squares of tissue, each about 5 mm x 5 mm. Place these in a dish of distilled water.

Mount one piece of tissue on a microscope slide in **distilled water** under a cover slip.

Mount the other pieces in **1 mol dm⁻³ potassium nitrate** solution.

Label your slides appropriately.

Examine the tissue mounted in distilled water, using your microscope. Find an area of the tissue where pigmented cells occur, preferably as a single layer.

- (a) Describe the distribution of the coloured contents within the cells. [1]

Observe the piece of tissue mounted in 1 mol dm⁻³ potassium nitrate solution, using your microscope.

- (b) (i) After about one minute, make a large drawing to show the detailed structure of **one** epidermal cell which is typical of the most deeply coloured cells which you can see.

On your drawing, label the positions of the:

- cell wall
- cell surface membrane.

[4]

- (ii) Account **fully** for the change in appearance of the cells when placed in 1 mol dm^{-3} potassium nitrate solution. [3]

Heavy metals such as lead and copper are toxic to plants.

- (c) Predict the appearance of the epidermal cells if the epidermis is mounted in 1 mol dm^{-3} **lead nitrate** solution. Explain your predictions. [3]

Fig. 2.1 shows the view of a mammalian white blood cell, using an eyepiece graticule and the high-power objective lens of a microscope.

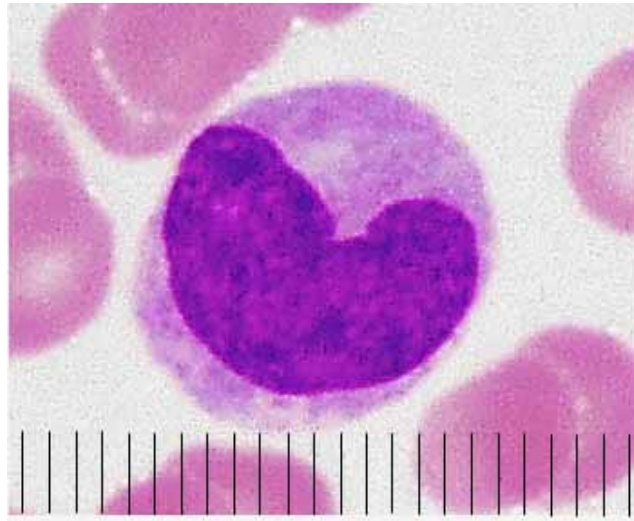


Fig. 2.1

- (d)** Use a table to record **three** observable differences between the blood cell in Fig. 2.1 and the cell you saw in **(a)**.

[3]

- (e) (i) The student calibrated the eyepiece graticule against a stage micrometer with the following results:

Number of eyepiece graticule divisions across 5 stage micrometer divisions = 25

One stage micrometer division = 0.01 mm

Use this information to calculate the actual length of the cell.

Show the steps and units in your calculation.

[3]

Actual length of the cell _____

- (ii) Calculate the magnification of the blood cell in Fig. 2.1.

Show your working.

[2]

Magnification _____

[Total: 19]

Question 3

Industrial wastewater contains high concentrations of fats, like oil and grease, which may pollute fresh water and influence aquatic environments. The use of lipase enzymes to remove oil and grease from the wastewater is an effective and environmentally-friendly treatment method. Chemicals are added to the wastewater to emulsify the fats and break them up into smaller fat droplets, increasing the surface area for lipase to break them down into fatty acids and glycerol at a higher rate.

In wastewater treatment, enzymes are used on a large scale. However, enzymes are costly and most are only commercially available in liquid or dehydrated forms. As such, once added to the wastewater treatment mixture, they cannot be recovered to be re-used, thus driving up the cost of treatment.

Researchers postulated that the immobilisation of lipase in alginate beads would allow the enzymes to be re-used, leading to cost savings. The enzymes can be added to sodium alginate and immobilised in the beads as shown in Fig. 3.1.

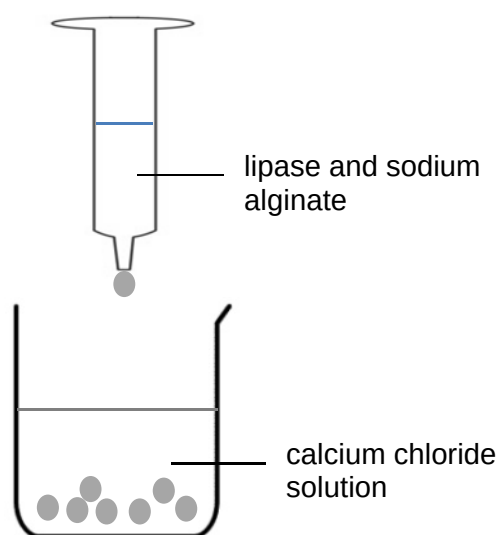


Fig. 3.1

Full-fat milk can be used as a substrate for the immobilised lipase enzymes. Bile salts act like detergents and can be used to emulsify the fats without affecting the pH. Sodium carbonate is a base that can be added to standardise the milk to an alkaline pH, prior to treatment with lipase.

The monitoring of the pH of the reaction mixture from a predetermined start to end-point pH can be used to measure the activity of the immobilised lipase.

Using this information and your own knowledge, design an experiment to find the concentration of lipase needed to form immobilised lipase beads with the same activity as free enzymes (non-immobilised).

You must use:

- 10% lipase solution
- industrial 3% lipase
- sodium alginate suspension
- calcium chloride solution
- full-fat milk
- sodium carbonate solution
- 5% bile salts solution
- a pH meter with a digital display to 2 decimal places
- stopwatch

You may select from the following apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rod, etc.
- syringes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

[illegible]

[illegible]

[illegible]